NeuroLens

Watch a brain watch your video.

PREDICTED BRAIN RESPONSE TO VIDEO, SECOND BY SECOND

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$10k+

to test one ad on a brain scanner

fMRI study	**$10 – 15k**	2–4 weeks
Facial coding	**$850 – 4,000**	3–5 days
AI attention maps	**$0.5–1.5k/mo**	sub-minute
NeuroLens	**$1.00**	≈ 7 min

Cost per advert or cut. fMRI vs surveys: Venkatraman et al., 2015. Competitor ranges: our report, section 1.3.
NeuroLens: $1.00 per 30 s video; about 7 minutes measured for a 30 s clip.

Ten cuts by lunch.
No way to judge them.

Generative tools made production cheap. The feed made every second count.



One upload in.
A brain response out.

Your video

MP4, MOV or WebM, up to two minutes

TRIBE v2

Meta's brain model, trained on real fMRI of people watching film

20,484 points

of the cortex, predicted every second

Three outputs

Timeline · 3D brain · AI report

For every second of the video. A virtual pre-test of a typical viewer, not a prediction of views or sales.

Two people, two questions.

“Which three seconds do I cut?”

A short-form creator uploads two edits of an intro, finds the dip, and re-cuts.

“Which variant gets the media budget?”

A small agency uploads up to five cuts, compares any pair, and decides with evidence.

Seven brain areas.
Three questions.

Seeing

faces · bodies · scenes · visual

Hearing

sound

Understanding

speech · social

BELOW TYPICAL

0 = TYPICAL FILM & TV VIEWING

ABOVE TYPICAL

From file to insight in five steps.

01
Sign in
Google or email · $10 invite credit

02
Upload
$1.00 per started 30 s, shown first

03
Predict
A GPU starts on demand

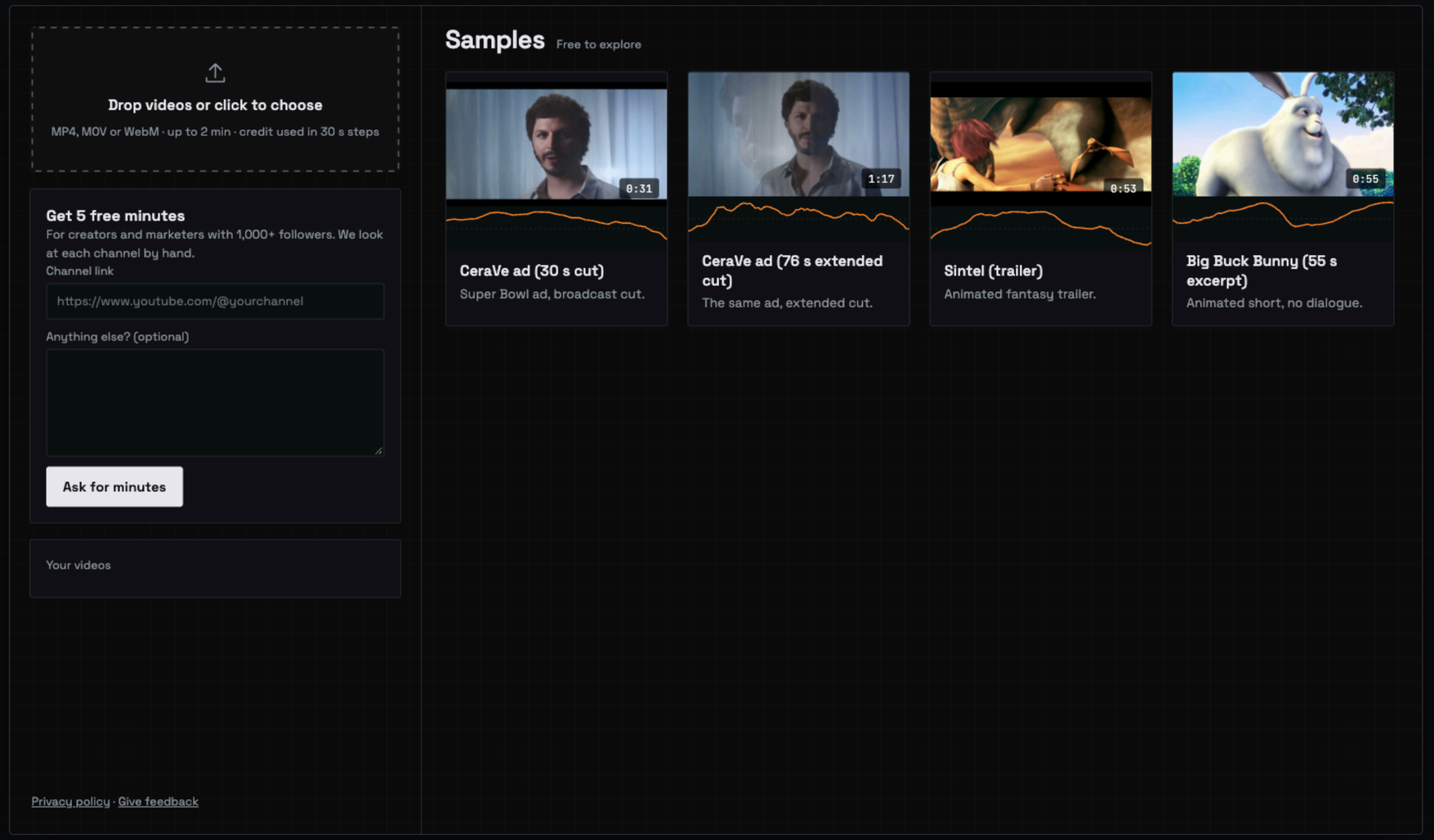
04
Watch
Live progress, stage by stage

05
Explore
Curve · 3D brain · report · compare

MP4, MOV, WEBM · UP TO 2 MINUTES · $1.00 PER STARTED 30 S OF VIDEO

LIVE

Sign in, upload, watch it process

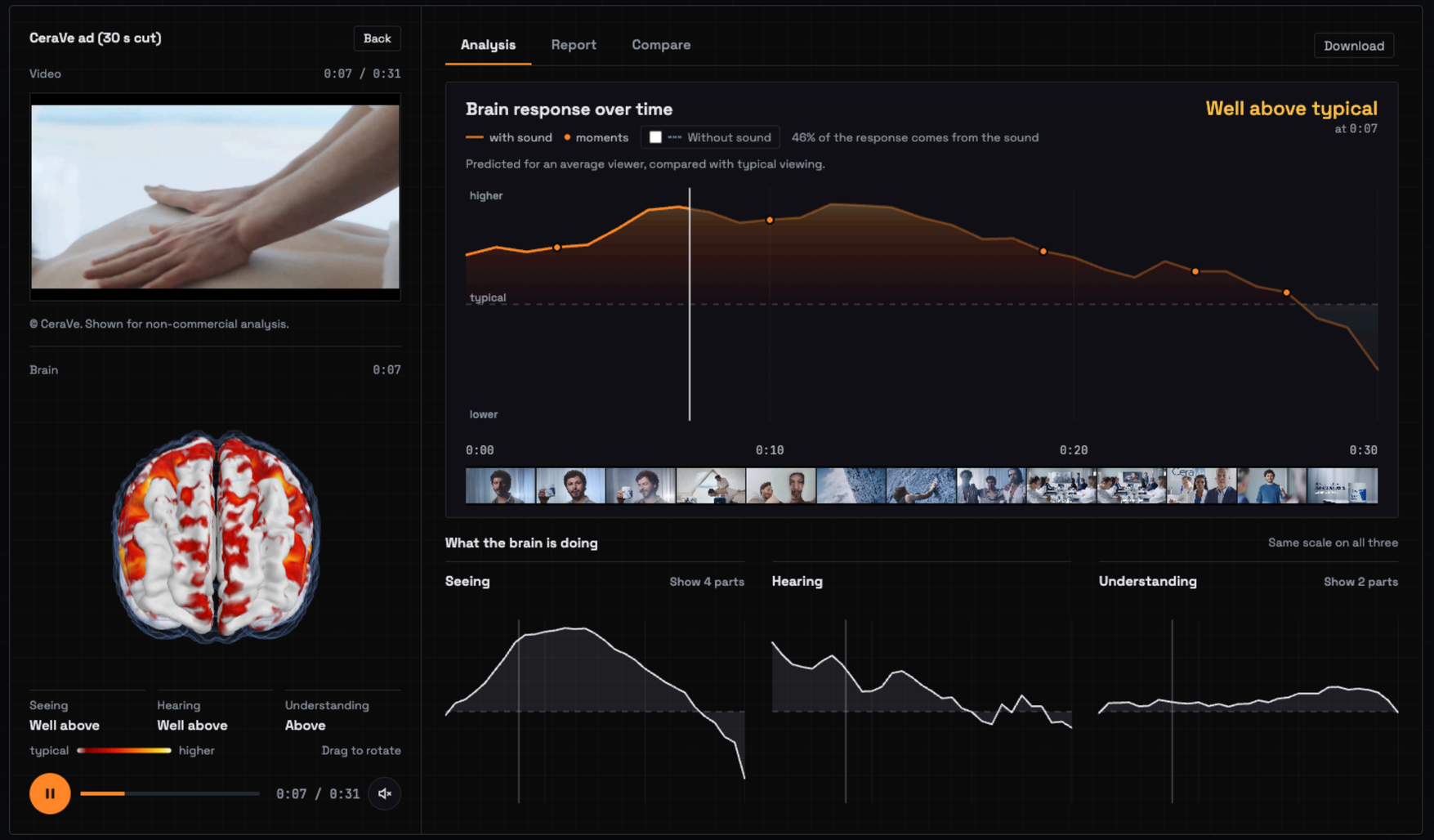


1. Sign in with Google or email
2. The price shows before you confirm
3. Stages tick: check, queue, analyse

THE UPLOAD SCREEN

LIVE

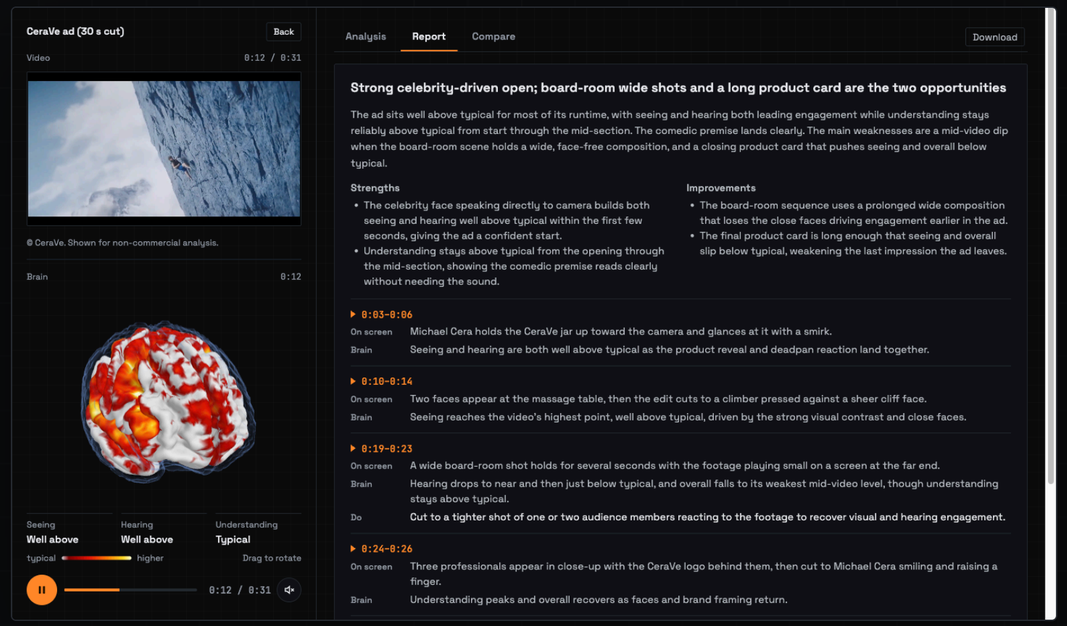
The results, second by second



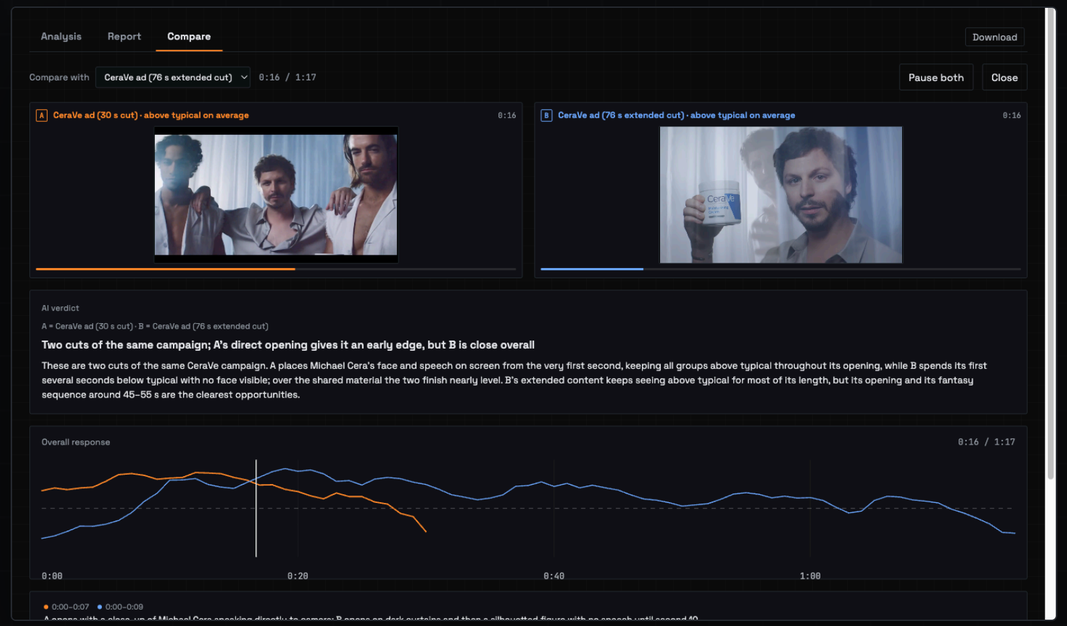
1. Follow the playhead
2. Peak near 0:06, dip at the end
3. Seeing · hearing · understanding
4. Toggle: without sound

● LIVE

The AI report, and comparing two cuts



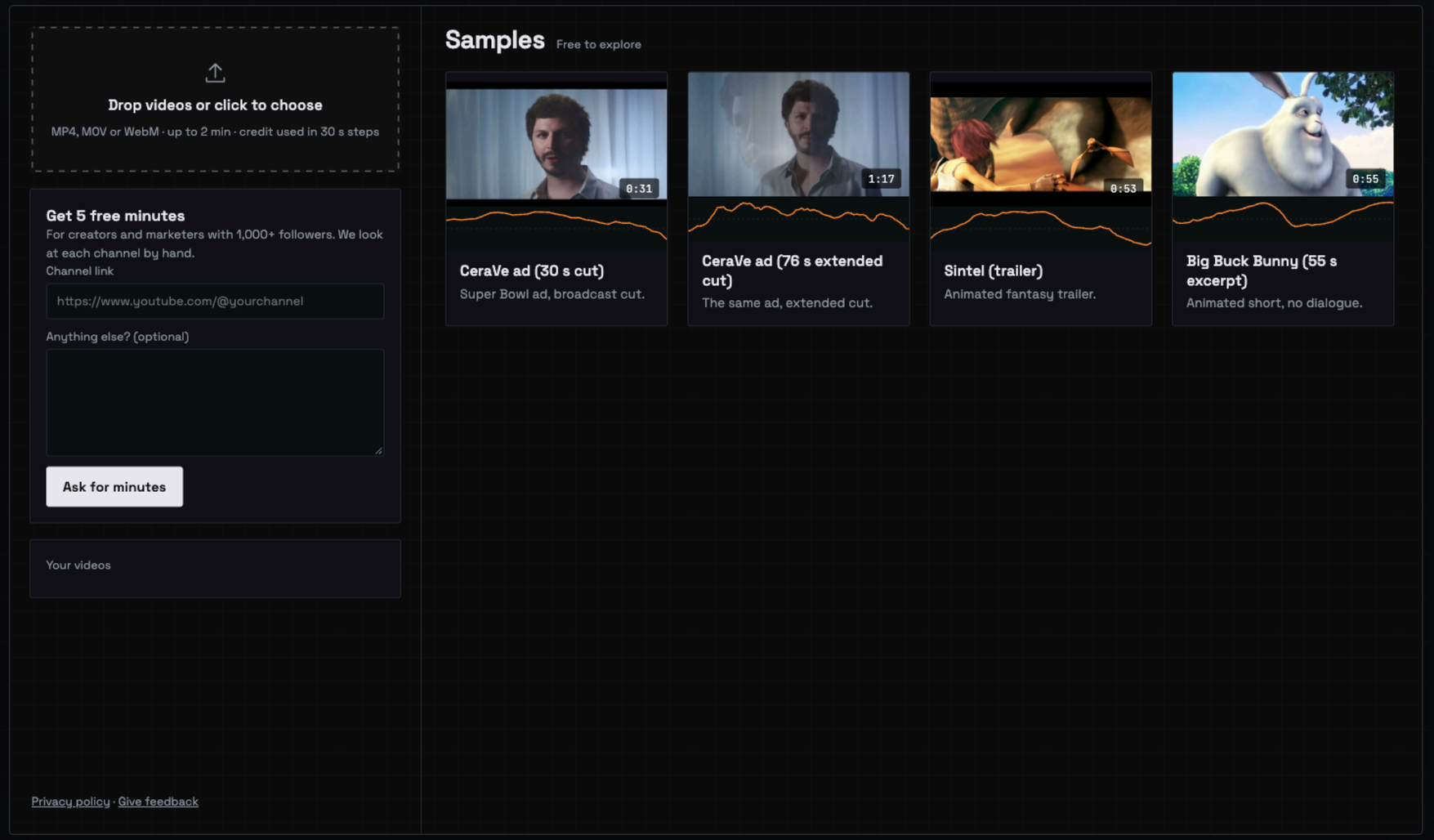
REPORT TAB



COMPARE TAB · 30 S CUT VS 76 S CUT

LIVE

A bad upload costs nothing



1. Upload a text file renamed .mp4
2. It is rejected within seconds
3. No credit is charged

REJECTED BEFORE ANY GPU STARTS

We pay for GPU minutes only when someone is waiting.



Check first

Length, size and credit are verified before any GPU wakes.

Scale to zero

Off after 7 idle minutes: about $5 a month when idle.

Refund by default

Credit is charged or refunded exactly once. A failed job costs nothing.

Pay as you go. Credit, not cards.

WHO AND HOW MUCH

$1.00 per started 30 s

For independent creators and small agencies picking one of many AI-made cuts. New users get $10 of credit: ten videos.

WHY PREPAID CREDIT

Card fees are fixed

About 2.9% + $0.30 a charge: a third of a $1.00 sale, but only 4% of a $20 pack.

Course version: an administrator grants credit; Stripe Checkout belongs to the production design. TRIBE v2 is CC BY-NC, so a paid launch needs a licence.

What one analysis costs us.

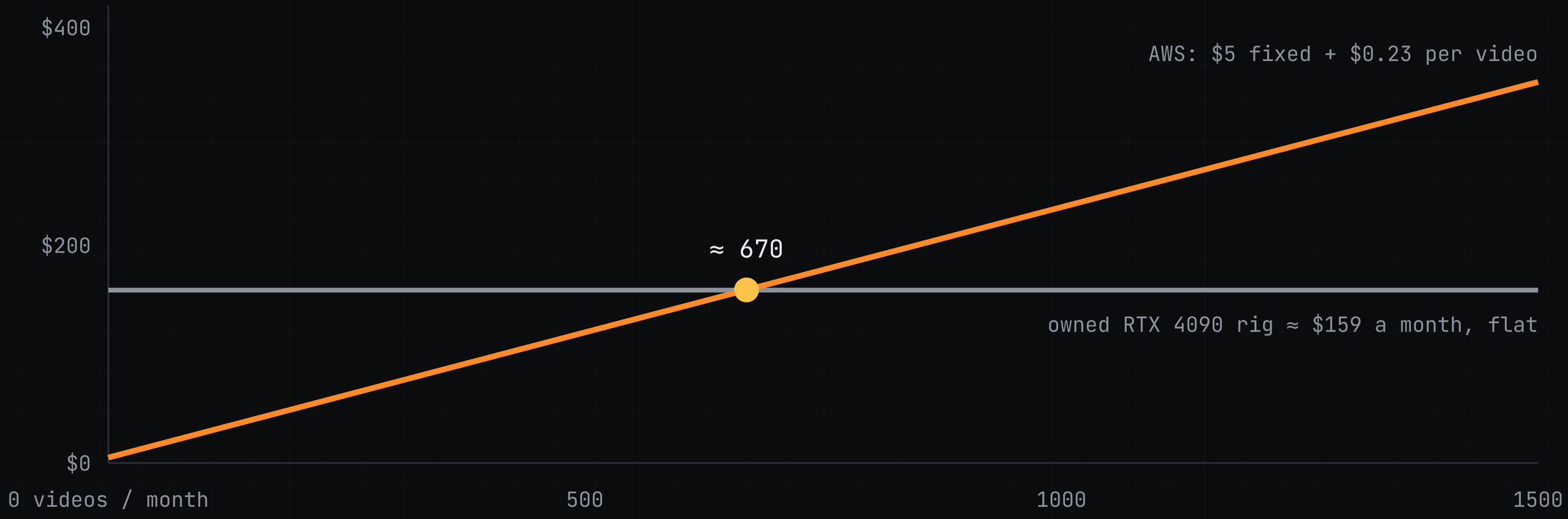
$0.14–0.23

our AWS cost per 30 s video, against a $1.00 price

GPU: 393 s at $1.22/h (L4)	$0.13
Start-up + idle, per session	7.5 + 7 min
Cost, 3 videos per session	$0.23
Cost, 30 videos per session	$0.14
Fixed cost per month	$5
Gross margin	**77 – 86%**

Report, Table 6. Cost per video = GPU time × hourly price + (start-up + idle) × hourly price ÷ videos per session. Spot GPUs (≈ $0.10 a video) were not available.

Cloud is cheaper below about 670 videos a month.



What it claims, and what it doesn’t.

WHAT IT IS

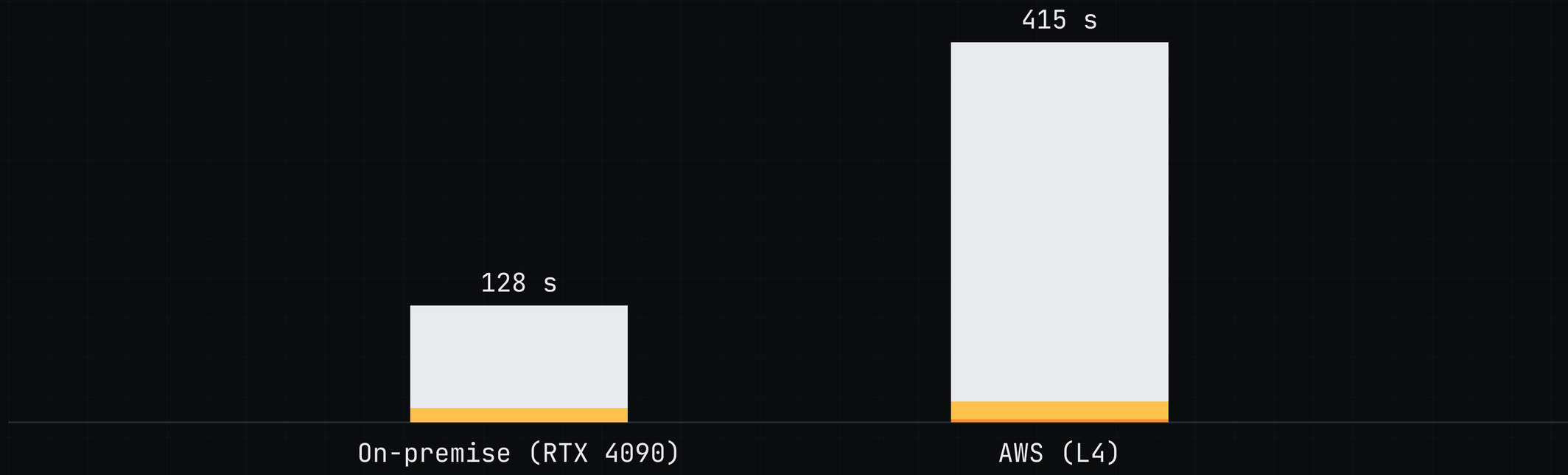
- An average viewer’s predicted brain response
- One scale for every video
- A virtual pre-test for editing

WHAT IT ISN'T

- A prediction of views or sales
- Instant: the first video after a quiet spell waits ≈ 6 min
- Cleared for commercial use (CC BY-NC)
- Statistically proven (small test)

Where does the time go in one job?

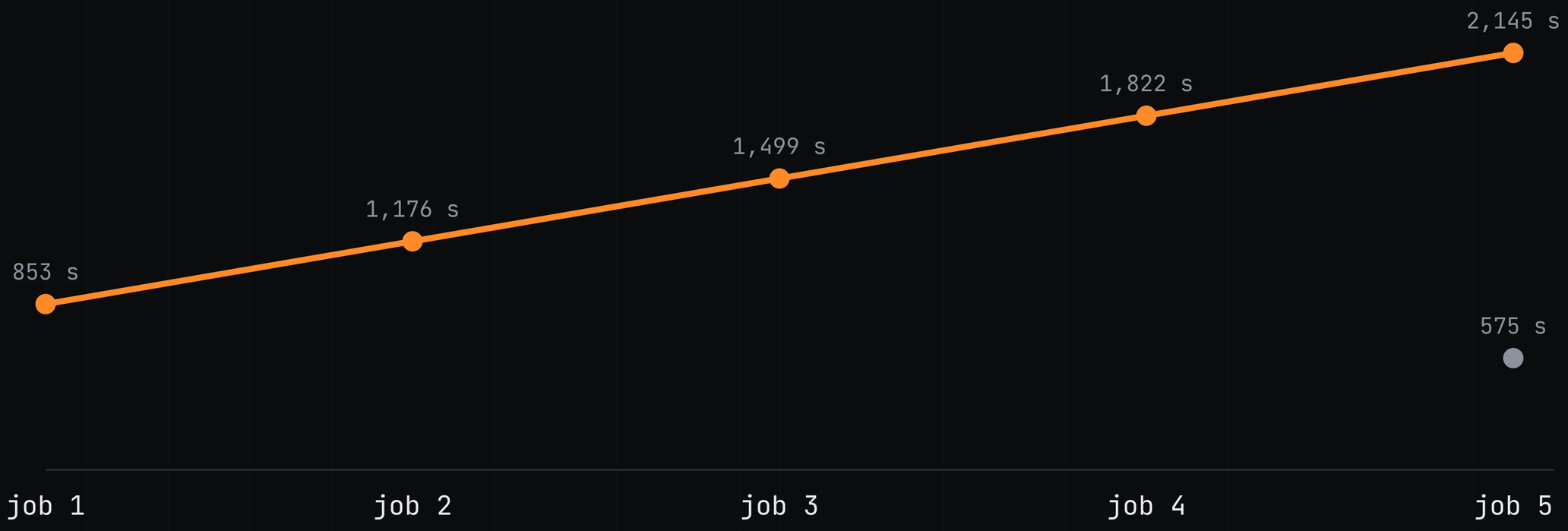
Same code, same 30 s clip, on our own RTX 4090 and on AWS.



GPU TIME PER VIDEO-SECOND: 3.71 S (RTX 4090) VS 10.7 S (A10G) · FIRST VIDEO AFTER 7+ MIN IDLE: 779 S ON AWS · SAME OUTPUT ON BOTH

Five people at once.

Five uploads at once. The endpoint is set to one GPU, so jobs run one after another.



SECONDS UNTIL EACH JOB FINISHES. NO JOB WAS DROPPED. IN 77 OF 80 AUTOMATED REFUND TESTS A FAILED JOB WAS REFUNDED EXACTLY ONCE.

Can people use it?

Five student creators. One task: upload an ad, find one peak and one drop-off.

[TBD] / 100

SUS SCORE · 68 IS AVERAGE · 80 IS AN A

COMPLETED UNAIDED, OF 5 · MEDIAN TIME ON TASK [TBD]

A $10,000 brain study, in 7 minutes for $1.

Live today: sign-in, upload, prediction, 3D brain, AI report, comparison, refunds.

Next

Licence

TRIBE v2 is CC BY-NC: a paid launch needs a licence.

Next

Scale

A GPU pool that grows with the queue: cheaper above about 1,000 videos a month.

Next

Polish

Fix what our usability study flagged.

Thank you.
Questions?

df6girqyykyfa.cloudfront.net

CODE · GITHUB.COM/KELVINLOWHM86/NEUROLENS

Recorded demo

[Embed the recorded demo of slides 8–11 here]

Beyond the proof of concept: what we learned.

Spot did not deliver

All 26 Spot launches failed for lack of capacity. Spot averaged only 1–9% below on-demand.

A smaller GPU fits

The model peaked at 19.9 GB, so a 24 GB L4 or A10G is enough.

The GPU sets the speed

119 s clip: 491 s on an L40S, 1,540 s on an L4.

Production design (EC2 GPU pool behind a load balancer): $0.10 – $0.20 a video, about $70 a month more fixed cost. Cheaper than SageMaker on the same L40S above about 480–1,440 videos a month. Not built: estimates.